

The empirical recurrence for OEIS A301840, recovered from its tail

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Abstract

OEIS A301840 records “Empirical recurrence of order 53” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 271$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 57$. Its last coefficient is $c_{53} = -96 \neq 0$, so no further tail satisfies anything shorter; any order-53 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A301840 is “Number of $n \times 7$ 0..1 arrays with every element equal to 0, 1 or 2 horizontally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

64, 2778, 72514, 1986213, 54948498, 1518341751, 41966406867, 1159968653556, ...

The entry states:

Empirical recurrence of order 53 (see link above)

As of the “Last modified” line on the live entry (Mar 27 2018 14:30:45, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 271$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 271$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 53: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 53.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 542$ exact terms from $n = 4$ on, it returns order 53 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{53} c_i a(n-i),$$

c_1	85	c_{15}	111618703512616	c_{29}	38079411203604784	c_{43}	1120416648590
c_2	−3152	c_{16}	−290120019221431	c_{30}	−32277407386097472	c_{44}	−231611906509
c_3	70815	c_{17}	687158222761848	c_{31}	25278657733786785	c_{45}	40473927751
c_4	−1113407	c_{18}	−1488740369371757	c_{32}	−18262339339676729	c_{46}	−5775523533
c_5	13298949	c_{19}	2960062130926924	c_{33}	12147887406648279	c_{47}	630037721
c_6	−127103614	c_{20}	−5416661626311547	c_{34}	−7424643648658231	c_{48}	−43949813
c_7	1006178523	c_{21}	9144022957745653	c_{35}	4159608054303457	c_{49}	217298
c_8	−6755523868	c_{22}	−14267404347585509	c_{36}	−2130409128682714	c_{50}	391974
c_9	39115802557	c_{23}	20605858360428549	c_{37}	994405813350184	c_{51}	−52072
c_{10}	−197704418233	c_{24}	−27575763053013109	c_{38}	−421485494272167	c_{52}	3368
c_{11}	880353318809	c_{25}	34216388694685746	c_{39}	161527829295590	c_{53}	−96
c_{12}	−3479466994810	c_{26}	−39375254589996969	c_{40}	−55677854860577		
c_{13}	12284593960995	c_{27}	42019394283807581	c_{41}	17149995293359		
c_{14}	−38965739507829	c_{28}	−41564042249092629	c_{42}	−4681661071012		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 57$, and it is the recurrence OEIS A301840 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{53} - c_1 x^{53-1} - \dots - c_{53}$. Its constant term is $-c_{53} = 96$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 53 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 53, so $\deg q = 53$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 53, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 53, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -96 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-53 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A301840.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.